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Executive Summary of the Thesis

Data Integration, Sensitivity Analysis and Inverse Problem Formulation for a Compartmental Colon Model

Laurea Magistrale in Mathematical Engineering - Ingegneria Matematica

Author: Michele Baggi

Advisor: Prof. Paola Francesca Antonietti

Co-advisors: Béatrice Laroche, Lorenzo Sala

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1. Introduction

The colon hosts a large community of bacteria that ferment the dietary fibre our own enzymes cannot break down. The same diet, however, does not give the same result in every person: with identical meals the metabolic response can differ by more than 100% from one subject to another. To explain this we would need to know what happens inside the colon of each individual, and that is what we cannot see: sampling the colon requires invasive procedures, so the only routine measurement is a stool sample, which tells us what leaves the colon and not what happens on the way.

This work aims to close that gap one sample at a time. On one side there is a mechanistic model of the colon, a nonlinear system of 106 ordinary differential equations that follows fibre from the moment it enters until it is excreted [2]. On the other there are 3613 faecal metagenomes, each summarised as a composition of four functional profiles that describe what a microbial community can do with fibre [1, 3]. The question is whether the model can be tuned, sample by sample, so that it reproduces the composition measured in that sample. If it can, an observation can be associated with a model that can then be simulated and perturbed.

The work proceeds as follows. The data and the reference model are presented in Section 2, and the model is adapted so that its microbial groups match the four functional profiles (Section 3). The model has sixty parameters, while one observed composition carries only three independent numbers, so most parameters are fixed and the influential ones are selected by global sensitivity analysis (Section 4). These are then estimated for each sample by Bayesian inversion, made feasible by a neural surrogate (Section 5). The work was carried out at MaIAGE (INRAE, Université Paris-Saclay) in the Dynenvie team.

2. Data and model

Sample data. I use the four functional profiles of Labarthe et al. [1]. Each faecal sample is read by shotgun sequencing, which reveals the genes present in the microbial community, which means the reactions it is equipped to carry out. Keeping only the genes involved in fibre and mucin degradation and counting together those performing the same reaction leaves 101 numbers per sample. On a reference cohort, Labarthe et al. decomposed these vectors once with the constrained non-negative matrix fac-

torisation of Raguideau et al. [3], learning four fixed non-negative combinations of the 101 entries, the functional profiles. Every sample is then described by four weights, one per profile: these coefficients $\boldsymbol{w} = (w_1, \dots, w_4)$ are the data used here.

The dataset contains 3724 samples from healthy subjects and several disease groups. Since the analysis below uses log-ratios, weights below 10^{-6} are treated as absent profiles and replaced by that threshold. Removing in addition the top 1 % of each of the three minor profiles leaves 3613 samples.

The model. The reference model [2] describes anaerobic carbohydrate degradation along the colon. The colon is split into three regions in series, and each region into a lumen and a mucus layer, giving six compartments (Figure 1). Each is assumed perfectly mixed, so every state follows a balance of what enters, what leaves and what reactions produce or consume. Lumen and mucus exchange solutes, bacteria and mucins, while the host secretes mucus and absorbs metabolites and water.

The same reactions run in every region. The sugar user x_1 hydrolyses fibre and ferments the sugars released; the lactate user x_2 and the hydrogen users x_3 and x_4 consume the products, with the last two producing acetate and methane respectively. All four groups also decay. Altogether the model can be written as

$$\dot{\boldsymbol{u}} = F(\boldsymbol{u}, \boldsymbol{\theta}, t), \qquad \boldsymbol{u} \in \mathbb{R}^{106},$$

with a discrete reset whenever the distal lumen reaches the excretion volume. This makes the system hybrid. I re-implemented in Python and reproduced the results of the original Matlab version.

The fractions f. For each microbial process, a fraction f of the consumed substrate is broken down into metabolic products through catabolism, while the remainder $1 - f$ is used for growth, becoming new biomass. A second set of fractions $\boldsymbol{\eta}$ plays the same role for the products of catabolism. The model is implemented directly in terms of these fractions, so every parameter set used below is chemically consistent by construction.

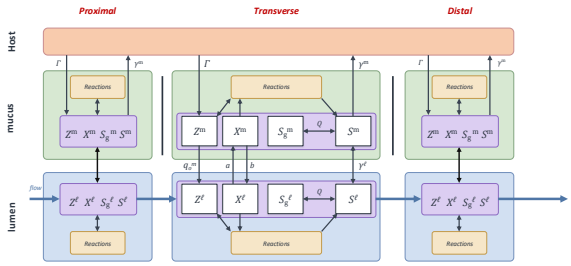


Figure 1: Structure of the model: three regions in series, each split into a lumen and a mucus compartment.

3. Model adaptation

The data give four profile weights, while the model gives four biomasses. Their absolute values cannot be compared, because the weights come from gene counts and have no common unit with biomass. For this reason we compare only their relative compositions:

$$\boldsymbol{y} = \frac{\boldsymbol{x}_D^l(t_e)}{\|\boldsymbol{x}_D^l(t_e)\|_1} \;\approx\; \boldsymbol{z} = \frac{\boldsymbol{w}}{\|\boldsymbol{w}\|_1}, \qquad (1)$$

where $\boldsymbol{x}_D^l(t_e)$ contains the four biomasses of the distal lumen at an excretion time. This is the main modelling assumption of the work: it links data and model, but it is not a direct measurement of biomass. It also assumes that functional potential is proportional to activity, supported by the recovery of the same profiles from expression data [1].

From specialists to generalists. Equation (1) asks each functional profile to correspond to one state of the model. The reference formulation is too rigid for this, because each microbial state is tied to one substrate. Profile 1 is close to x_1 and Profile 4 to x_4, but Profiles 2 and 3 take part in several pathways, so we let them participate in more than one reaction, with fixed weights $\boldsymbol{\alpha} = (0.5, 0.3, 0.2)$ and $\boldsymbol{\beta} = (0.8, 0.2)$, set once and shared by every sample: any variability between individuals is left entirely to the parameters estimated later, and Section 4 checks that this fixed choice has little influence on the output. The kinetic laws are unchanged. Since the four uptake rates are linear in biomass, the

effective rates are

$$\begin{aligned} \tilde{\rho}_1 &= \rho_1(x_1), & \tilde{\rho}_2 &= \rho_2(x_1 + \alpha_1 x_2), \\ \tilde{\rho}_3 &= \rho_3(\alpha_2 x_2 + \beta_1 x_3), & \tilde{\rho}_4 &= \rho_4(\alpha_3 x_2 + \beta_2 x_3), \\ \tilde{\rho}_5 &= \rho_5(x_4). \end{aligned} \tag{2}$$

The biomass balances are modified accordingly, so that each group grows only from the share of the reactions it carries. Setting $\boldsymbol{\alpha} = (0, 1, 0)$ and $\boldsymbol{\beta} = (0, 1)$ recovers the reference model exactly and was used as an implementation test.

Two further changes. The bacterial inflow fixes the smallest abundance the model can represent. The reference values place this floor above some compositions in the dataset, so the inflow was reduced and assigned almost entirely to the first group. The adherence coefficient, which is zero in the reference model, was also given a small positive value so that a mucus population can be restored from the lumen.

Stopping rule and robustness. Runs are stopped using the four distal biomasses recorded at excretion. For each biomass we sum its absolute changes over a 30-day window, divide by its current value, and take the maximum over the four groups. The run stops when this quantity falls below 10^{-3}. This avoids declaring convergence only because the system is moving slowly. Dependence on the initial state cannot be studied analytically here, so every state variable was independently perturbed by $\pm 50\,\%$ for 1000 initial conditions. At the final excretion, the spread of each distal biomass is reduced by about three orders of magnitude and that of the composition by about five. This is numerical evidence, not a proof, but it supports the use of a common initial condition in the rest of the work.

4. Sensitivity analysis

The model has many parameters, but only some can change the distal composition. I identify them in two steps. Morris screening ranks many factors at low cost by changing one factor at a time along trajectories. Sobol' indices [4] are then used on the retained factors to decompose output variance. The first-order index S_i measures the contribution of θ_i alone, while the total index

$$S_{T_i} = \frac{\mathrm{E}[\mathrm{Var}[Y \mid \boldsymbol{\theta}_{\sim i}]]}{\mathrm{Var}[Y]} = S_i + \sum_{j \neq i} S_{ij} + \cdots \tag{3}$$

collects every contribution involving θ_i. The difference between S_{T_i} and S_i measures interactions.

Setup. The output is the composition $\boldsymbol{y}$ of the four distal biomasses in Equation (1). Its components sum to one, so only three are independent.

Sensitivity indices depend on the parameter ranges. Since sample-specific ranges are not available, positive parameters are varied log-uniformly between one half and twice their nominal value. Compositional parameters are perturbed in log coordinates and then renormalised. The only exception is the decay rate, varied over one decade because it represents several loss processes not described explicitly. Half-saturation constants, volumes, pH and conversion factors are fixed. This gives 30 factors.

Results. The Morris screening used 60 trajectories, corresponding to 1860 simulations. At the bottom of the ranking are the transport coefficients of the fermentation acids, mucin secretion, adhesion and the $\boldsymbol{\alpha}$ factors. Fibre intake is also weak: it strongly changes the total biomass but not its relative composition. At the top are the catabolic fractions f, the glucose split $\boldsymbol{\eta}$ and the decay rates; these twelve factors are retained.

Sobol' analysis was then run on these twelve factors with $N = 2048$, corresponding to 53 248 simulations (Figure 2). The f block carries most of the variance: the sums of its first-order indices are 0.91, 0.65, 0.70 and 0.74 on the four fractions. The other two blocks are much smaller. The decay rates never exceed $S_T = 0.07$, despite their wider range, and the largest index of the glucose split is 0.18. Interactions are weak except for the smallest fraction y_4.

Choice of the parameters. Influence alone is not enough: a parameter may strongly affect the output but still be almost constant between individuals. The catabolic fractions are a plausible source of inter-individual variability, since each microbial group aggregates different microorganisms that may split their substrate between catabolism and growth differently. The four fractions also act on different parts of the

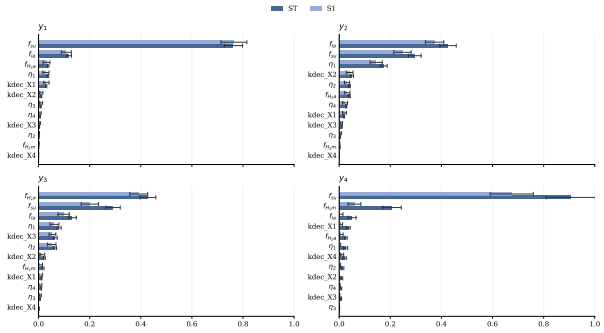


Figure 2: First-order (S_1) and total (S_T) Sobol' indices of the twelve retained factors, on the largest distal fraction y_1 and the smallest one y_4. $N = 2048$, $k = 12$, i.e. 53 248 simulations.

output. Hence I estimate

$$\boldsymbol{\theta} = (f_1, f_2, f_3, f_4).$$

A four-part composition has only three degrees of freedom, so four parameters cannot in general be identified uniquely from one observation. This implies there is at least one direction that the data alone cannot determine. Moreover, Sobol' indices are global averages: they justify the choice of the block but do not prove local identifiability for one sample.

5. Inverse problem

The forward problem maps parameters to a composition; the inverse problem estimates parameters from an observed composition. Since the parameter vector is not identifiable from the data alone, we use a Bayesian formulation. The prior regularises directions that are weakly constrained by the likelihood, while the posterior describes the remaining uncertainty. The goal is not to recover a unique biological parameter vector, but to find model configurations that reproduce the observed composition.

The posterior is sampled by Markov chain Monte Carlo using DEMetropolisZ [5], which builds proposals from differences between states visited earlier and adapts to the shape of the posterior without manual tuning.

Setup. Only the four catabolic fractions are varied. By construction $0 < f < 1$; numerically we use

$$\boldsymbol{\theta} \in [0.01, 0.99]^4$$

considering almost the whole admissible range, excluding only the boundaries to avoid degenerate cases.

Model and data are compositions, so Euclidean distance on the raw fractions is not appropriate. For a positive four-part composition $\boldsymbol{v}$ define

$$\operatorname{clr}(\boldsymbol{v})_i = \log v_i - \frac{1}{4}\sum_{j=1}^{4} \log v_j,$$

and the Aitchison distance

$$d_a(\boldsymbol{u}, \boldsymbol{v}) = \| \operatorname{clr}(\boldsymbol{u}) - \operatorname{clr}(\boldsymbol{v}) \|_2.$$

The likelihood is

$$\log \pi(\boldsymbol{z} \mid \boldsymbol{\theta}) \propto -\frac{1}{2\sigma^2} d_a^2\left(\widehat{G}(\boldsymbol{\theta}), \boldsymbol{z}\right), \qquad \sigma = 0.2, \tag{4}$$

where $\boldsymbol{z}$ is the observed composition and $\widehat{G}$ is the surrogate defined below. The priors are independent truncated Gaussians on $[0.01, 0.99]$, with standard deviation 0.25 and centred on the reference values of [2].

Surrogate. One simulation of the full model takes several seconds, so using it inside every MCMC step for every sample would be too expensive. I trained a neural surrogate of the forward map on $2^{15} = 32\,768$ parameter vectors generated with a scrambled Sobol' sequence and evaluated with the full model.

The surrogate is a multilayer perceptron with four hidden layers of 128 units and a final softmax, so its output is a composition by construction. It is trained using squared Aitchison distance. On the test set, the median Aitchison error is 0.012, the 95th percentile is 0.060, and R^2 is above 0.998 in clr coordinates.

One sample. The complete procedure was first tested on the medoid of the 3613 cleaned samples, the real observation with the smallest average Aitchison distance from all the others. Six chains reach $\widehat{R} \leq 1.002$ with more than 7800 effective samples. All four observed fractions fall inside the 95 % posterior predictive intervals (Figure 3), and the predictive median differs from the observation by less than 6 % on every component. Different parameter combinations give almost the same reconstruction, as expected from the underdetermined problem.

The whole dataset. The same procedure was applied independently to all 3613 compositions, using shorter chains. The calculation took 2 h 46 min on 64 cores and every case ran to completion.

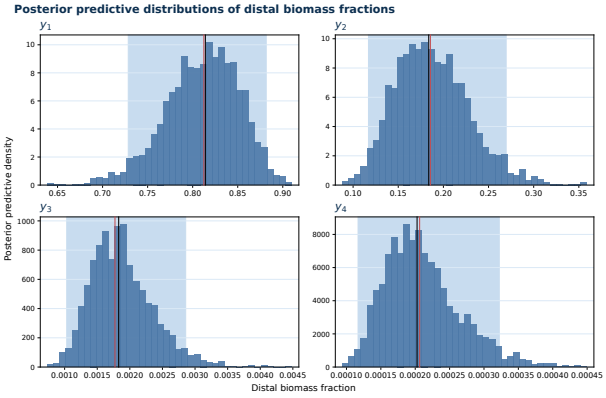


Figure 3: Posterior predictive distribution of the four distal fractions for the representative sample. Shaded region: 95 % interval. Black line: predictive median. Red line: observation.

samples	mean	median	max
100	0.166	0.142	0.429

Table 1: Aitchison distance between the data and the full mechanistic model, re-simulated at the posterior median, on 100 randomly selected samples.

The model–data misfit was assessed on 100 randomly selected samples: for each one, the posterior median of the four parameters was re-simulated with the full mechanistic model, giving a median Aitchison distance of 0.142. For comparison, the observed samples have an average distance of 3.24 from their compositional centre, so the reconstructed compositions lie an order of magnitude closer to the data than the natural variability between subjects.

Interpretation. The fit quality does not make the estimated fractions identifiable: a four-part composition carries only three independent numbers, so many parameter vectors reproduce the same observation, and the posterior stays close to the prior, because there is not enough information to move it. The fractions obtained for a sample should therefore not be read as that individual's true biological values. What the procedure does provide is a model configuration compatible with the observed composition, one among the many available; this is what makes it possible to simulate or perturb the model to study quantities that a stool sample alone cannot reveal.

6. Conclusions

The gap motivating this work was that identical diets can produce very different metabolic responses, and the only routine measurement, a stool sample, does not say why. Fitting the model's catabolic fractions to each sample's functional composition closes part of that gap: across the full cohort, the reconstructed compositions matched the data well within the natural variability between subjects, confirmed with the full mechanistic model.

This is not a measurement of an individual's true parameters: a four-part composition constrains only three degrees of freedom, and the profile-to-state correspondence of Section 3 is itself a modelling choice. What the inversion gives is a configuration compatible with the data, sufficient to simulate quantities the stool does not measure, such as short-chain fatty acid production, but not to claim a unique biological identity for the sample.

To get more meaningful results, this approach would need richer information, in different ways. The functional profiles compress the metagenomic reads into four numbers, and a richer representation would keep information that is now lost, helping the model fit each sample better. New measurements, such as short-chain fatty acids, total biomass, repeated samples, or detailed diet records, would also provide additional constraints. In particular, richer dietary information would make it possible to apply the same workflow to existing models that distinguish between different macronutrients.

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